SPL). However, the ear cannot distinguish between sounds that are closer than about 1 dB (12% in amplitude) apart. In other words, there are only about 120 different loudness levels that can be detected, spaced logarithmically over an amplitude range of one-million.

This is important for digitizing audio signals. If the quantization levels are equally spaced, 12 bits must be used to obtain telephone quality speech. However, only 8 bits are required if the quantization levels are made *unequal*, matching the characteristics of human hearing. This is quite intuitive: if the signal is small, the levels need to be very close together; if the signal is large, a larger spacing can be used.

Companding can be carried out in three ways: (1) run the analog signal through a nonlinear circuit before reaching a linear 8 bit ADC, (2) use an 8 bit ADC that internally has unequally spaced steps, or (3) use a linear 12 bit ADC followed by a digital look-up table (12 bits in, 8 bits out). Each of these three options requires the same nonlinearity, just in a different place: an analog circuit, an ADC, or a digital circuit.

Two nearly identical standards are used for companding curves: **μ255 law** (also called **mu law**), used in North America, and **"A" law**, used in Europe. Both use a logarithmic nonlinearity, since this is what converts the spacing detectable by the human ear into a linear spacing. In equation form, the curves used in μ255 law and "A" law are given by:

EQUATION 22-1
Mu law companding. This equation provides the nonlinearity for μ255 law companding. The constant, μ, has a value of 255, accounting for the name of this standard.

$$y = \frac{\ln(1+\mu x)}{\ln(1+\mu)} \qquad \text{for } 0 \le x \le 1$$

EQUATION 22-2
"A" law companding. The constant, *A*, has a value of 87.6.

$$y = \frac{1+\ln(Ax)}{1+\ln(A)} \qquad \text{for } 1/A \le x \le 1$$

$$y = \frac{Ax}{1+\ln(A)} \qquad \text{for } 0 \le x \le 1/A$$

Figure 22-7 graphs these equations for the input variable, *x*, being between -1 and +1, resulting in the output variable also assuming values between -1 and +1. Equations 22-1 and 22-2 only handle positive input values; portions of the curves for negative input values are found from symmetry. As shown in (a), the curves for μ255 law and "A" law are nearly identical. The only significant difference is near the origin, shown in (b), where μ255 law is a smooth curve, and "A" law switches to a straight line.

Producing a stable nonlinearity is a difficult task for analog electronics. One method is to use the logarithmic relationship between current and